

Setup

System A: distilbert-base-uncased, fine tuned using the organizers' starter kit ([main.py](#)). Two seeds, identical hyperparameters otherwise. Trained on the official train split only. The final epoch is reported for both seeds. Two DeBERTa-v3-base runs were attempted first but they failed to learn; training loss came to around 1.39 which is equivalent to guessing each label $\frac{1}{4}$ times. **System B: Qwen2.5-1.5B-Instruct**, used. Two prompt variants were used zero-shot and few-shot (same instructions as zero shot but with 4 worked examples, one per label taken from the train split). Model outputs were parsed by checking whether one of the 4 valid label strings appeared in the generated text.

Results

Representative Model	Architecture	Weighted F1	SoftCons	HardCons
DeBERTa v3-base (Pilot)	Encoder	0.75	0.84	0.79
OPT-125M (Pilot)	Decoder	0.61	0.65	0.58
BART-large (Pilot)	Encoder-Decoder	0.68	0.80	0.72
A. DistilBERT (mean)	Encoder	0.4349	0.3755	0.2708
B.Qwen (Few-shot)	Decoder	0.4194	0.3357	0.2238
B.Qwen (Zero-shot)	Decoder	0.2025	0.2130	0.0325

Table 1: Pilot Results, System A: fine-tuned encoder and System B: prompted LLM

Analysis

1. How do Systems A and B compare on each of the three metrics? How do they compare with the pilot results in the task README?

System A has a higher weighted F1 than System B (both cases) and also higher SoftCons and HardCons. Systems A and B both have much lower values than the Pilot results on all three metrics.

2. Which labels does each system confuse most? Include the confusion matrix from the scorer.

True/Pred	BWD	EQV	FWD	NEG
BWD	132	28	26	4
EQV	43	104	18	9
FWD	117	30	40	3
NEG	45	27	19	15

Table 2: Confusion Matrix for System A (Seed 42)

True/Pred	BWD	EQV	FWD	NEG
BWD	82	25	80	3
EQV	23	110	34	7
FWD	54	28	104	4
NEG	35	20	34	17

Table 3: Confusion Matrix for System A (Seed 43)

True/Pred	BWD	EQV	FWD	NEG
BWD	57	63	29	41
EQV	30	104	29	11
FWD	16	55	74	45
NEG	23	23	15	45

Table 4: Confusion Matrix for System B (Few Shot)

True/Pred	BWD	EQV	FWD	NEG
BWD	32	1	132	25
EQV	54	2	106	12
FWD	59	0	104	27
NEG	43	0	37	26

Table 5: Confusion Matrix for System B (Zero Shot)

System A confused Backward and Forward the most in both seeds, but the direction of the bias differed. In Seed 42 (Table 2), the model over-predicted BACKWARD, 117 out of 190 true FORWARD pairs were assigned BACKWARD, while only 26 of the 190 BACKWARD pairs were assigned FORWARD. It predicted BACKWARD 337 times total against 190 true BACKWARD pairs. For seed 43 (Table 3), the directional confusion was more balanced, 80 of 190 true BACKWARD pairs were called FORWARD and 54 of 190 true FORWARD pairs were assigned BACKWARD. This system somewhat reliably detects that there is an entailment relation between the pairs, but fails to assign the directions accurately, it differs based on the random seed.

System B Few Shot predicted EQUIVALENT the most (245 times, only 104 correct), likely its fallback when unsure of direction. It was not as confused about directional entailment (BACKWARD called FORWARD only 29 times; FORWARD called BACKWARD 16 times), suggesting that it more often fails to detect a direction at all, rather than confusing two directions; falling back to EQUIVALENT or NEGATIVE instead (NEGATIVE assigned 142 times against only 106 true cases). **System B Zero Shot** predicted FORWARD most often (379 times against 190 True). And it rarely predicted EQUIVALENT. Directional confusion was also high: BACKWARD called FORWARD 132 times, FORWARD called BACKWARD 59 times. Across both variants of System B, directional confusion was quite prevalent, consistent with System A.

3. Pick 10 errors from your best system and describe any patterns you see (for example, word overlap, negation, quantifiers such as "all" and "some").

text1	text2	true_label	predicted_label
inside a cow-schemed , fur-covered dome	inside and black and white furry enclosure	FORWARD	EQUIVALENCE
as Manager	Ferguson	NEGATIVE_OTHER	EQUIVALENCE
with a ball	with a tennis ball	BACKWARD	FORWARD
from a tree	in a tree	NEGATIVE_OTHER	FORWARD
with network-sharing deal	network-sharing , spectrum deal	NEGATIVE_OTHER	FORWARD
Mubarak release	Mubarak 's release request	BACKWARD	EQUIVALENCE
in an orange dress	in a dress	FORWARD	BACKWARD
US stocks	Stocks	FORWARD	BACKWARD
A small orange and white dog	A brownish and white dog	FORWARD	NEGATIVE_OTHER
fresh election for president	fresh presidential election	EQUIVALENCE	BACKWARD

A pattern of directional reversal can be seen, see above highlighted in yellow, the model does correctly identify an entailment but predicts the opposite direction, this is in line with the confusion matrix. As seen in the green highlighted pairs, where the phrases share a lot of vocabulary, the model seems to read that as an EQUIVALENCE relationship, even when the true relationship is weaker or they are unrelated entirely.

4. Find 5 pairs where your system is inconsistent under reversal (fails SoftCons). What does the model seem to be doing wrong?

Data from System A, Seed 43

first_text1/second_text2	first_text2/second_text1	first_gold	first_pred	second_gold	second_pred
's war shrine visit	's shrine visit	FORWARD	BACKWARD	BACKWARD	BACKWARD
anti-protest laws	anti-protest law	FORWARD	BACKWARD	BACKWARD	BACKWARD
Iran nuclear experts and also world powers	World powers and Iran	FORWARD	FORWARD	BACKWARD	FORWARD
of all Rage rallies	march of anger	FORWARD	BACKWARD	BACKWARD	BACKWARD
Security video	Video	FORWARD	BACKWARD	BACKWARD	BACKWARD

4 of these 5 examples share the same structural error, the model predicts the identical directional label (BACKWARD) for both the original and flipped versions of the same pair, and the fifth example predicts FORWARD for both directions. Together these suggest the model is not evaluating each direction independently, instead it appears to pick up some feature of the pair as a whole and default to the same directional label regardless of which phrase comes first.

5. Write one idea you would test next to improve consistency, and why you expect it to help. You do not need to implement it.

Both the confusion matrices and SoftCons failure point towards the same issue, the model is not judging each direction of a pair on its own, in 4 out of 5 of the SoftCons failures it gave the exact same label, regardless of whether the pair was flipped or original, this suggests that it may be reacting to something about the pair as a whole rather than comparing the phrases. Perhaps during training, the model should process each pair's two directions together in the same batch and add a second term to the loss that directly penalizes the model when its two predictions contradict each other in the reverse versions. This should directly raise SoftCons, since this penalty is built to reward it.